

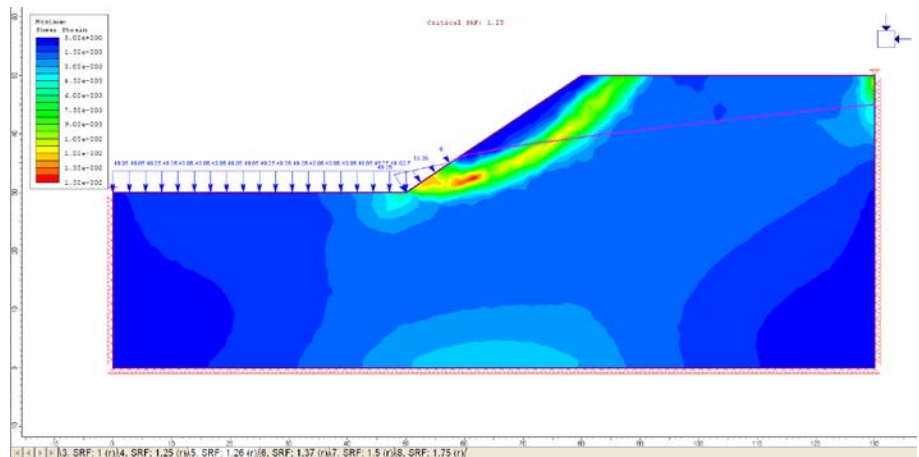
Drawdown Analysis for Slope

In this tutorial, two different methods of modeling the staged drawdown of ponded water against a slope are examined, and an SSR slope stability analysis is performed. The drawdown is implemented in two ways: 1) by lowering the water table, and 2) by running a finite element groundwater analysis with changing boundary conditions.

The complete models can be found in the **Tutorial 26 Slope Drawdown (piezo).fez** and **Tutorial 26 Slope Drawdown (finite).fez** files which can be accessed by selecting File > Recent Folders > Tutorials Folder from the *Phase2* main menu.

Topics covered

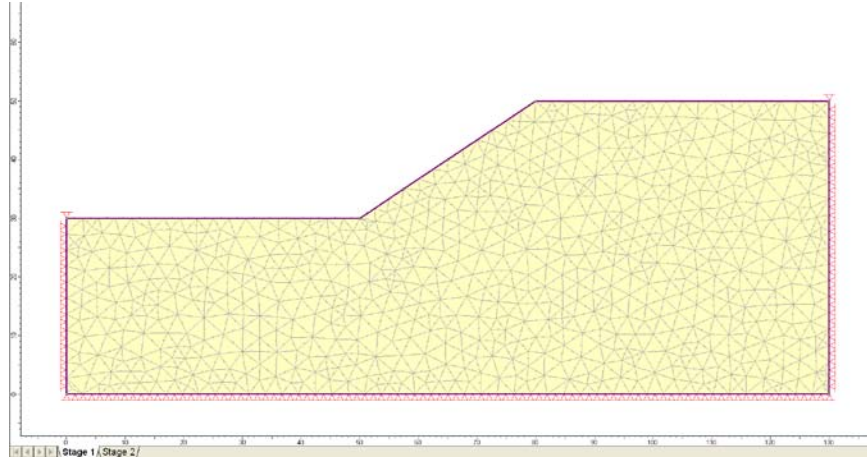
- Drawdown
- Staged piezo lines
- Staged groundwater analysis
- Shear strength reduction
- Importing coordinates
- Ponded water loading



Model



Start the *Phase2* Model program. In this tutorial we will start with a model that has already been constructed. Select **File** → **Open** and choose **Tutorial 26 Slope Drawdown (initial).fez** from the Examples > Tutorials folder in your *Phase2* installation folder. You will see a model that looks like this:

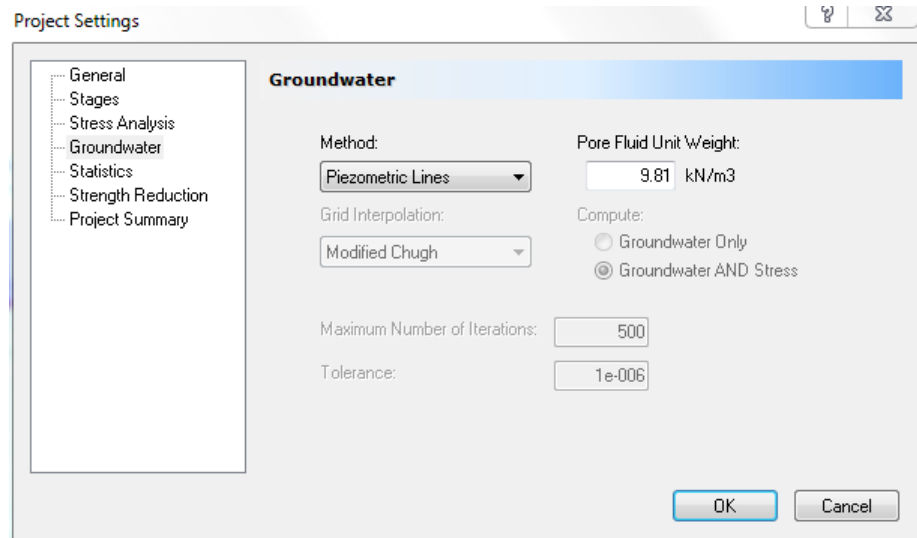


This is a two-stage model of a slope upon which a Shear Strength Reduction analysis will be performed. The geometry, material properties and mesh have already been specified for this model. The purpose of this tutorial is to examine how to simulate groundwater drawdown. This can be done in two different ways: using piezo lines or finite element groundwater analysis. We will start with the piezo line approach.

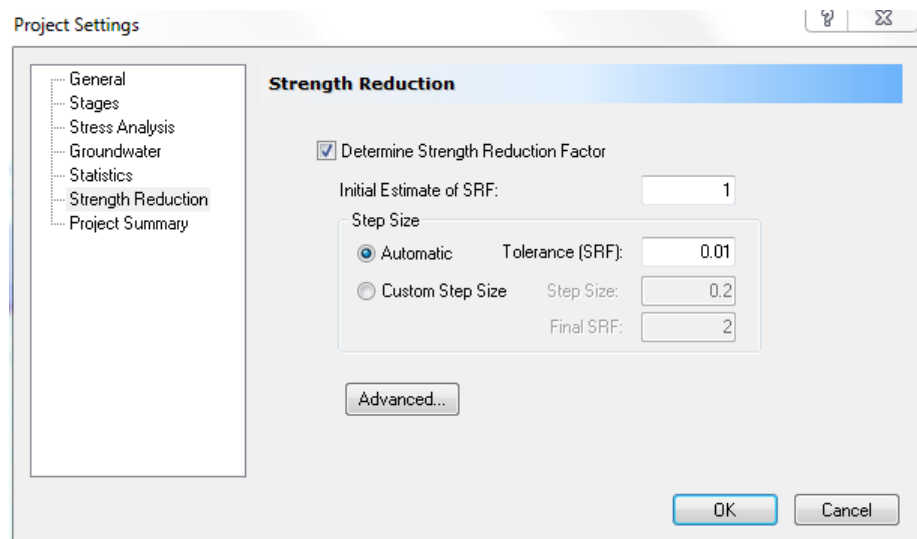
Drawdown with piezometric lines



Go to **Analysis** → **Project Settings** and select the Groundwater tab. Ensure that the Method is set to Piezometric Lines and that the Pore Fluid Unit Weight is 9.81 kN/m^3 as shown:



Click on the Strength Reduction tab and you will see that Shear Strength Reduction has been turned on.



Click OK to close the dialog.

Piezometric Lines

In the first stage, there will be 10 m of ponded water at the base of the slope. In the second stage, the ponded water will drop down 5 m.



Ensure you are looking at Stage 1. Go to **Boundaries** → **Add Piezometric Line**. Enter the following coordinates:

0 , 40

65 , 40

66.35 , 40.6

67.7 , 41

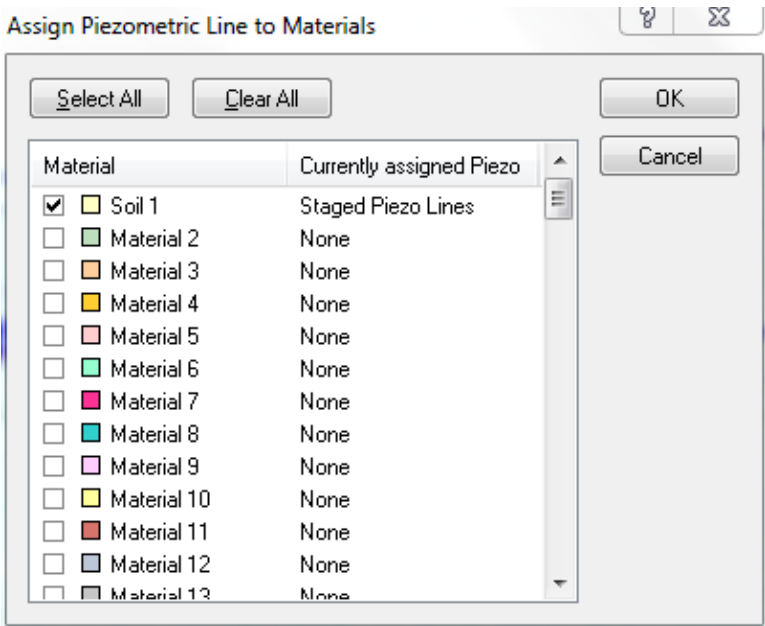
78.3 , 42.4

89.4 , 43.5

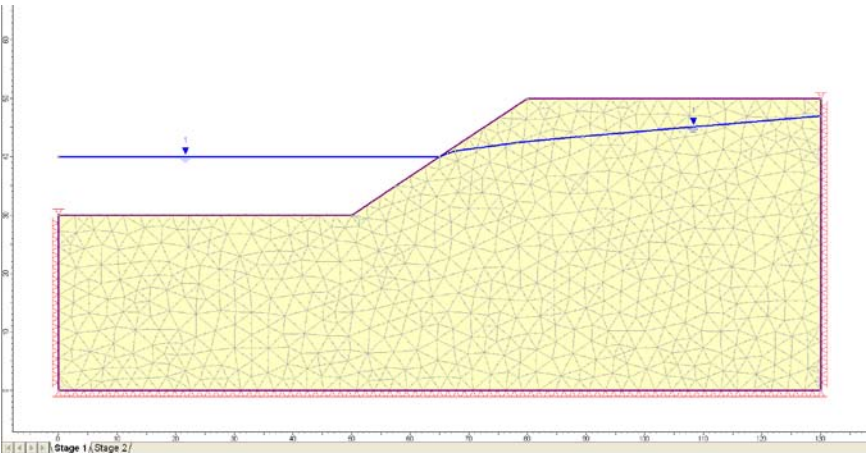
130 , 47

Enter

Now select Soil 1 in the dialog as shown:



Click OK. Your model should now look like this:





In the second state we want to drop the ponded water by 5 m. Click on the tab for Stage 2. Go to **Boundaries** → **Add Piezometric Line**. Enter the following coordinates (or import *Tutorial 26 piezo 2.txt* as described in the tip above):

0 35

57.5 35

58.7 35.8

60.1 36.3

65.3 37.3

76.6 39

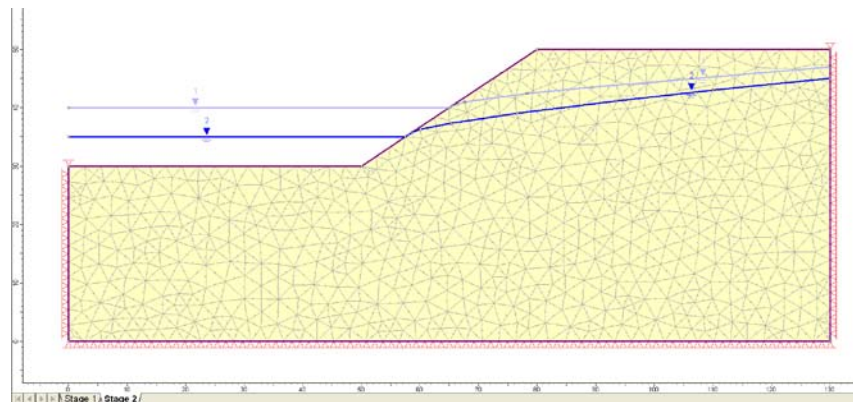
93.5 41.1

110 43

130 45

Enter

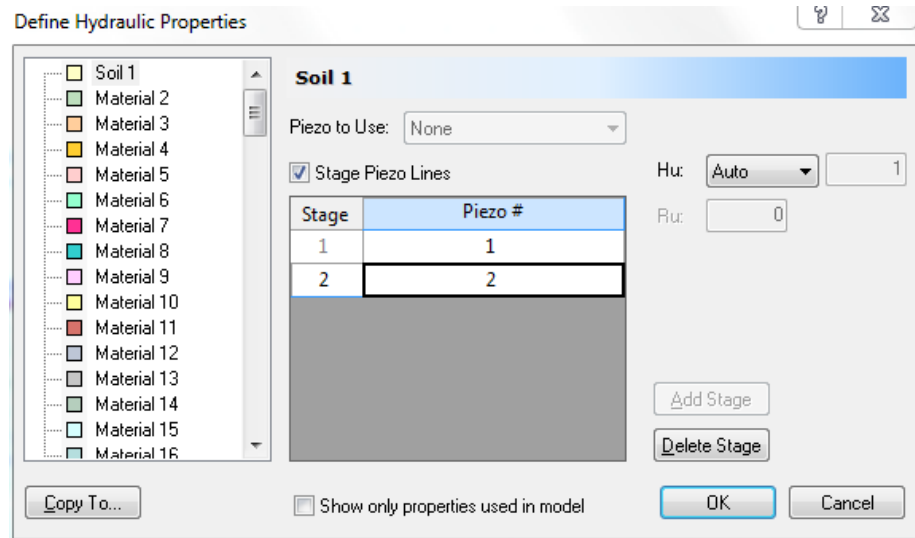
Select Soil 1 in the 'Assign Piezometric Line to Materials' dialog and click OK. The model should now look like this:



Notice that the new water table (piezo 2) has superseded the old water table (piezo 1). Go back to Stage 1. You will see that Stage 1 is the same as Stage 2; the active water table is piezo 2, and piezo 1 is inactive. We now need to stage the piezo lines.



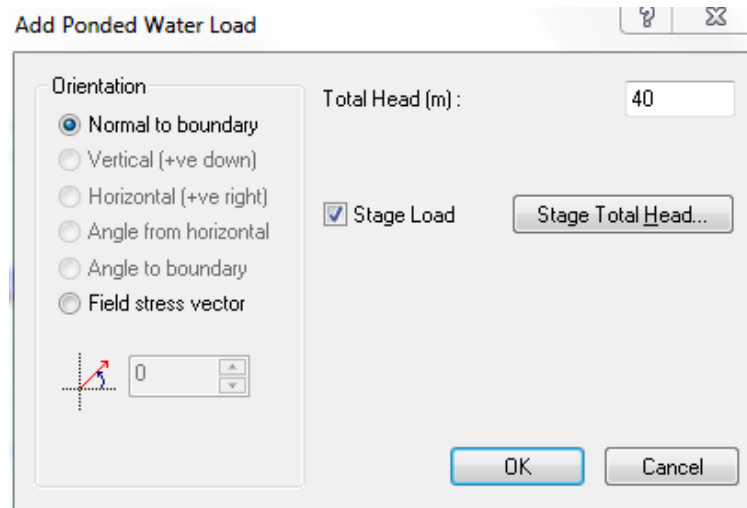
From the **Properties** menu, choose **Define Hydraulic**. Click on the Stage Piezo Lines checkbox. For Stage 1, choose Piezo #1 and for Stage 2, choose Piezo #2.



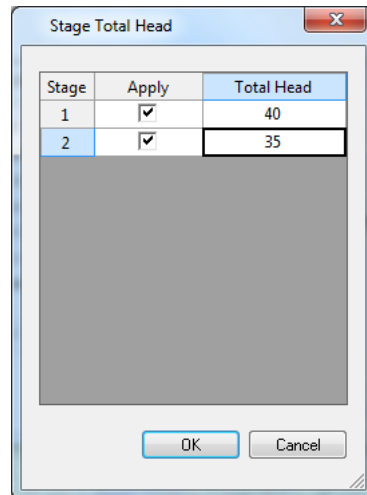
Click OK to close the dialog. Now click through the stages and you will see that piezo 1 is active in Stage 1 and piezo 2 is active in Stage 2.

Ponded Water Load

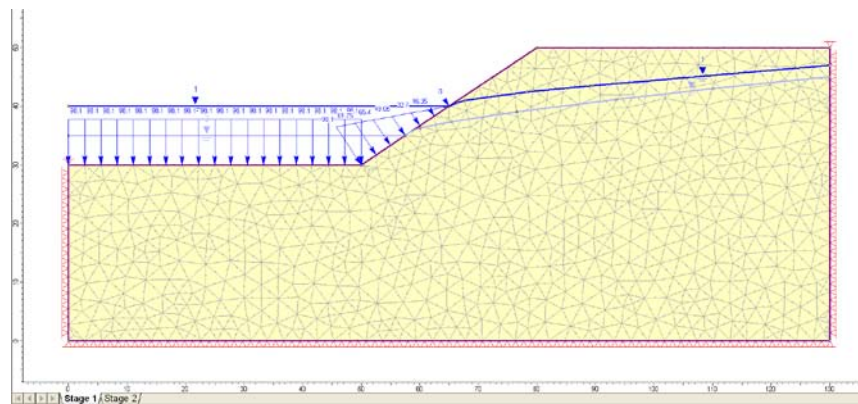
In this model there is ponded water at the left side of the model. This applies a stress to the soil surface due to the weight of the water. To account for this, go to **Loading** → **Distributed Loads** → **Add Ponded Water Load**. In Stage 1, the y-coordinate of the water table is 40 m. So enter a Total Head of 40 m. Now select the Stage Load checkbox.



Click the Stage Total Head button. In Stage 2, the water table is at 35 m, so change the Total Head for Stage 2 to 35 m as shown:



Click OK and then click OK to close the 'Add Pondered Water Load' dialog. You will now be prompted to select the boundary segments on which to apply the pondered water load. Select the horizontal surface between (0,30) and (50,30). Also select the two segments on the slope below piezo 1. Hit Enter to finish choosing segments. The model for Stage 1 should look like this:



You can see the load due to the weight of the water in Stage 1 ranges from 0 to 98.1 kPa. In Stage 2, the depth of pondered water is reduced by half so the maximum load is 49.05 kPa.



The model definition is now complete. Save the model using the **Save As** option in the **File** menu.

Compute



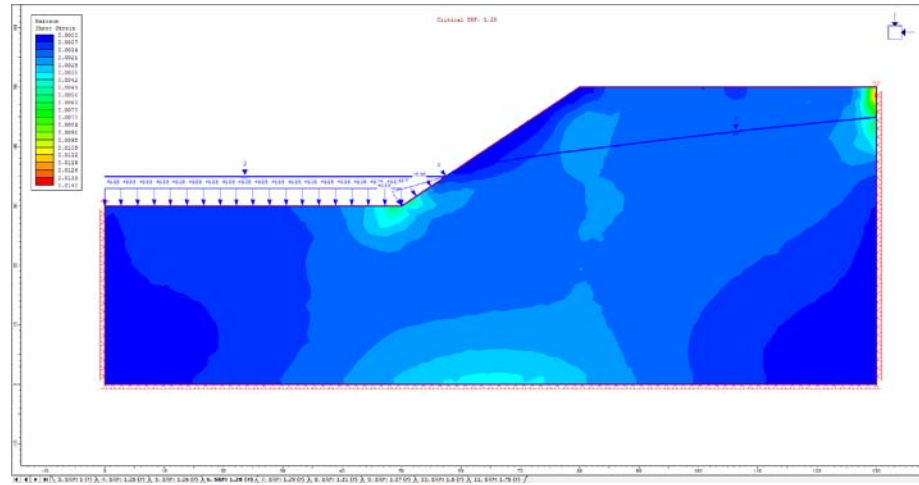
Run the model using the **Compute** option in the **Analysis** menu. Because it is performing a Shear Strength Reduction analysis, the model will take several minutes to run.



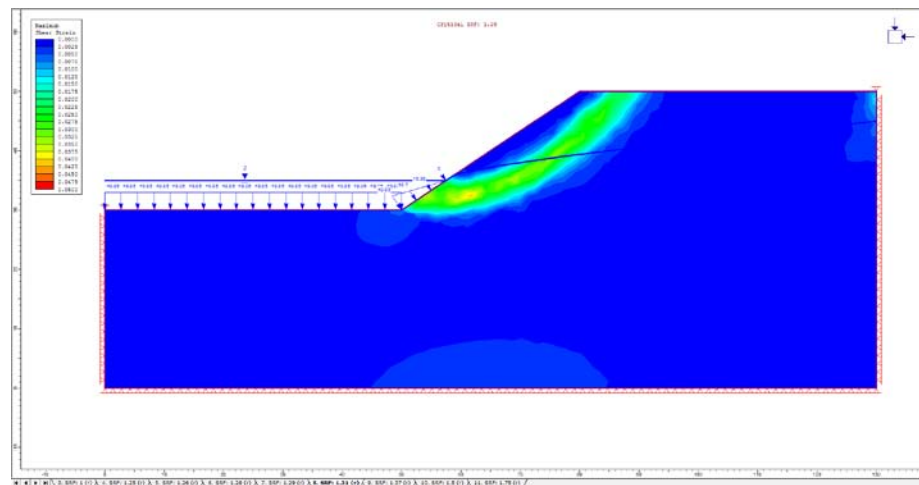
Once the model has finished computing (Compute dialog closes), select the **Interpret** option in the **Analysis** menu to view the results.

Interpret

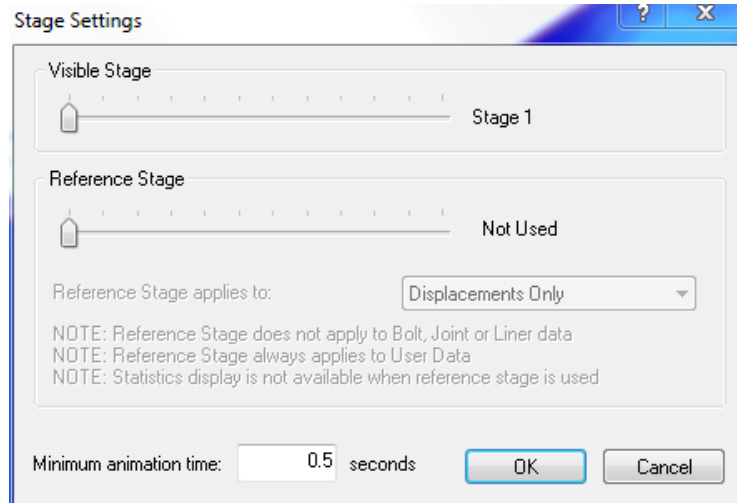
The Interpret program starts and reads the results of the analysis. The Shear Strength Reduction analysis is only performed on the **last stage of your model**, so what you are seeing is the maximum shear strain for the critical strength reduction factor (1.28) for the second stage.



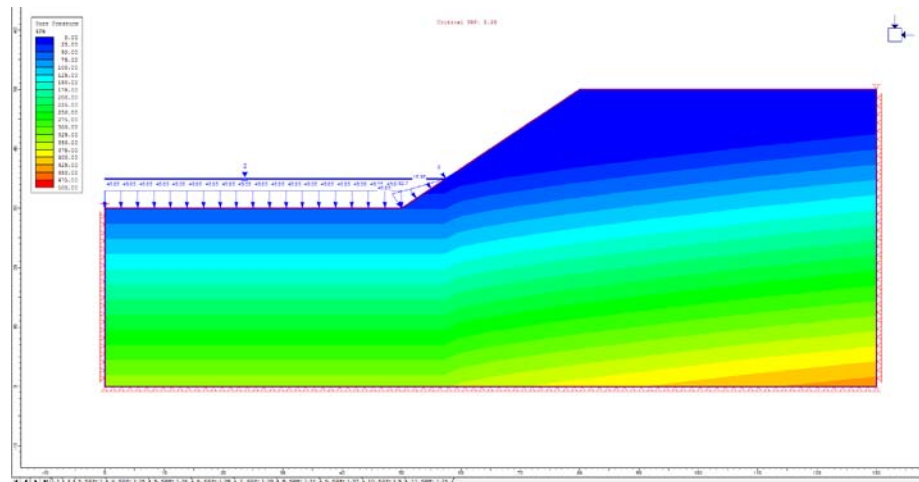
If you click the tab for SRF: 1.31 you get a better picture of the critical failure surface as shown.



If you want to look at the stage data prior to the SSR analysis, select **Stage Settings** from the **Data** menu. Set the Reference Stage to Not Used, and the Visible Stage to Stage 1.



Click OK. You will now see the maximum shear strain in stage 1. Change the plot to Pore Pressure using the pull-down menu at the top. You can see the pore pressure due to the water table in Stage 1. Click the tab for Stage 2. It is clear how the pore pressure decreases as the water table is lowered.

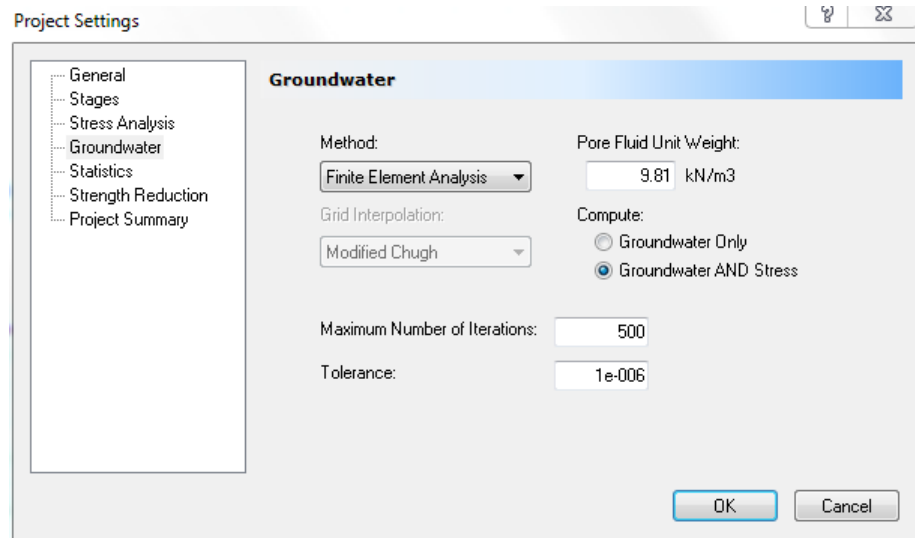


Drawdown with finite element groundwater analysis

We can perform the same analysis using finite elements to compute the pore pressures in the model instead of Piezometric lines.



Go back to the *Phase2* Model program. Go to **Analysis** → **Project Settings** and select the Groundwater tab. Change the method to Finite Element Analysis. Leave all other options as shown:

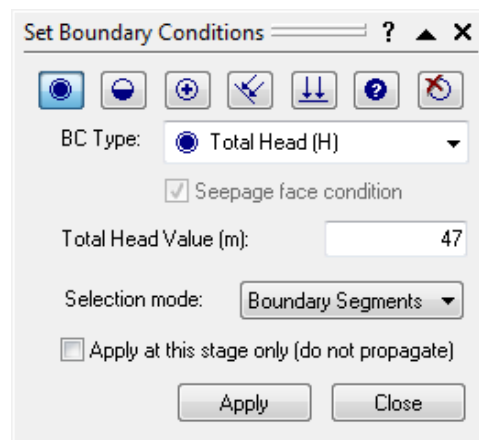


Click OK to close the dialog. *Phase2* will ask you if you want to delete the Piezometric lines. Click Yes.

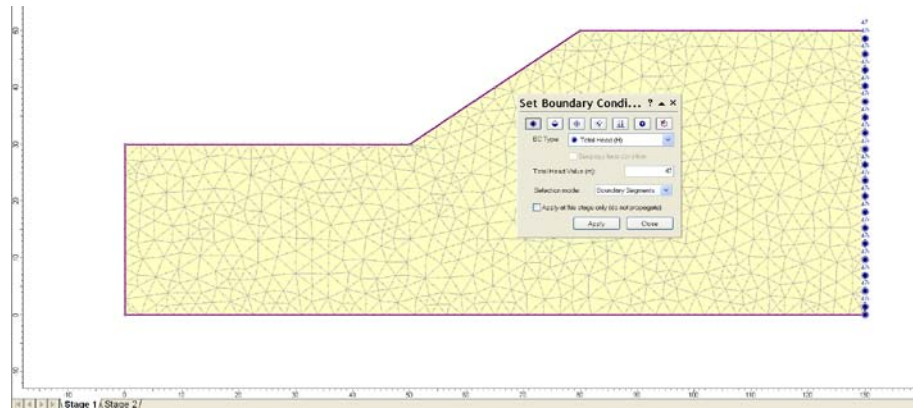
Groundwater boundary conditions



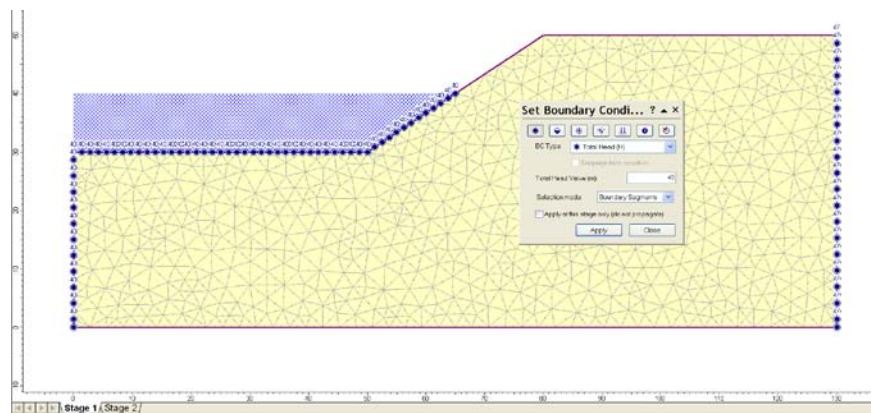
From the **Groundwater** menu, select **Show Boundary Conditions**. Ensure you are looking at Stage 1. Choose **Set Boundary Conditions** from the **Groundwater** menu. For the BC Type, choose Total Head. Set the Total Head Value to 47 m.



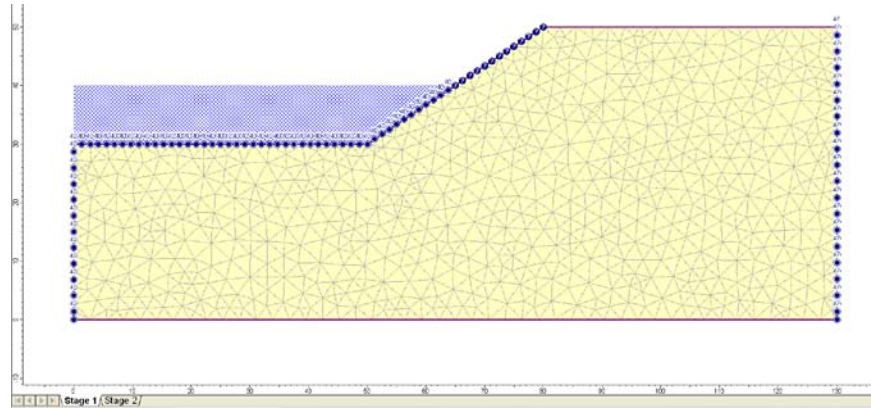
Use the mouse to select the right vertical boundary of the model. Click Apply in the dialog. The right side of the model should display a total head boundary condition as shown:



In the dialog, change the Total Head Value to 40. Click on the left vertical boundary, the horizontal segment at the base of the slope and the bottom two segments of the slope and click apply. The model should look like this:



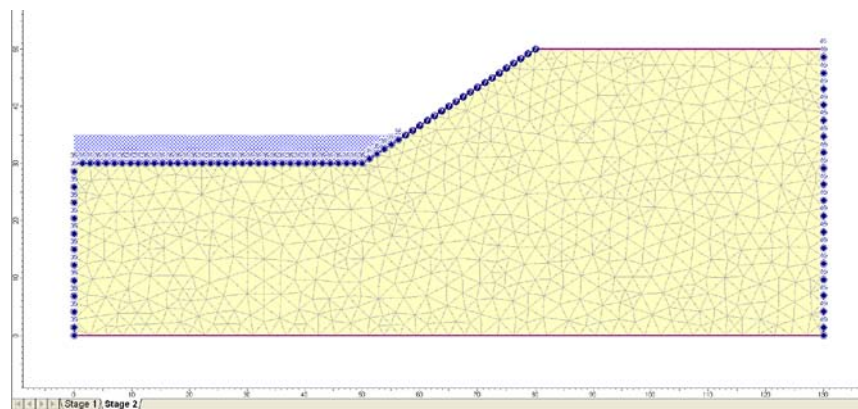
You can see how *Phase2* displays the ponded water based on the specified total head boundary conditions. Finally, we need to set unknown boundary conditions for the rest of the slope face, since we don't know where the water table will intersect the slope. In the Set Boundary Conditions dialog, select Unknown ($P=0$ or $Q=0$) for the BC Type. Click the top segment of the slope face and click Apply. Click the Close button and the model should look like this:



Now click the tab for Stage 2. In Stage 2 we want to lower the ponded water. Choose **Set Boundary Conditions** from the **Groundwater** menu again. Follow these steps:

- Select Total Head (H) for the BC Type. Set the Total Head to 45 m. Select the right vertical boundary and click Apply.
- Set the Total Head Value to 35 m. Click on the left vertical boundary, the horizontal boundary at the base of the slope, and the bottom section of the slope face and click Apply.
- Change the BC Type to Unknown ($P=0$ or $Q=0$). Click on the section of the slope just above the water table (the middle section of the slope face) and click Apply.

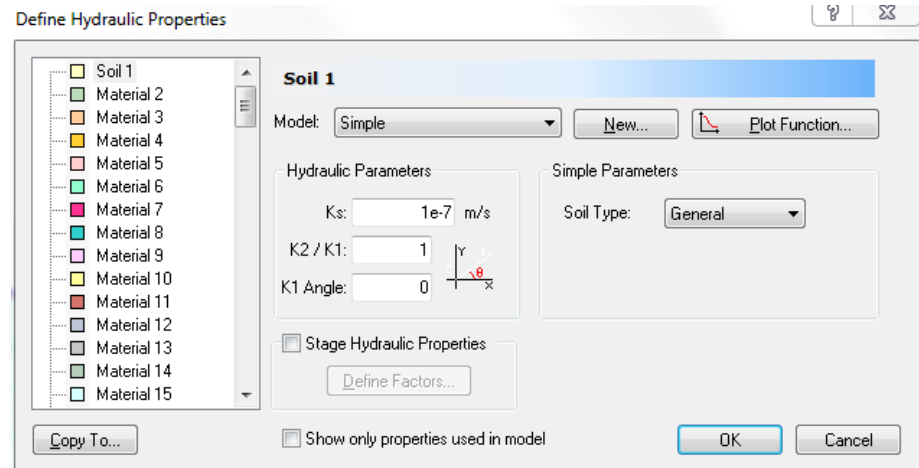
Select the Close button to close the dialog. The model should look like this for Stage 2.



Hydraulic material properties



From the **Properties** menu, choose **Define Hydraulic**. Here you can select the model that dictates the permeability transition from saturated to unsaturated soil. Leave the model as the default (Simple). You can also change the permeability here. Leave the default value of $1e-7$ m/s.



Click OK to close the dialog.

The model definition is now complete. Save the model using the **Save As** option in the **File** menu. Choose a different name from the model that you created using Piezometric lines.

Compute



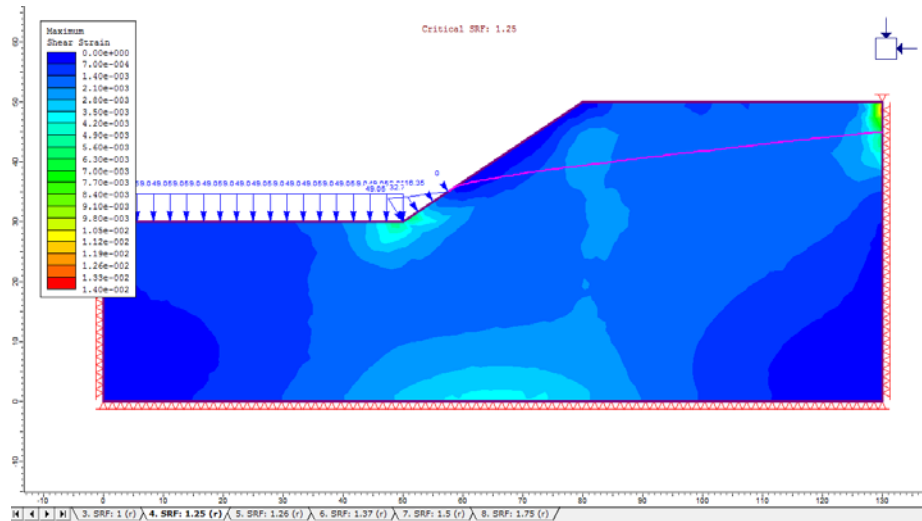
Run the model using the **Compute** option in the **Analysis** menu. Because it is performing a Shear Strength Reduction analysis, the model will take several minutes to run.



Once the model has finished computing (Compute dialog closes), select the **Interpret** option in the **Analysis** menu to view the results.

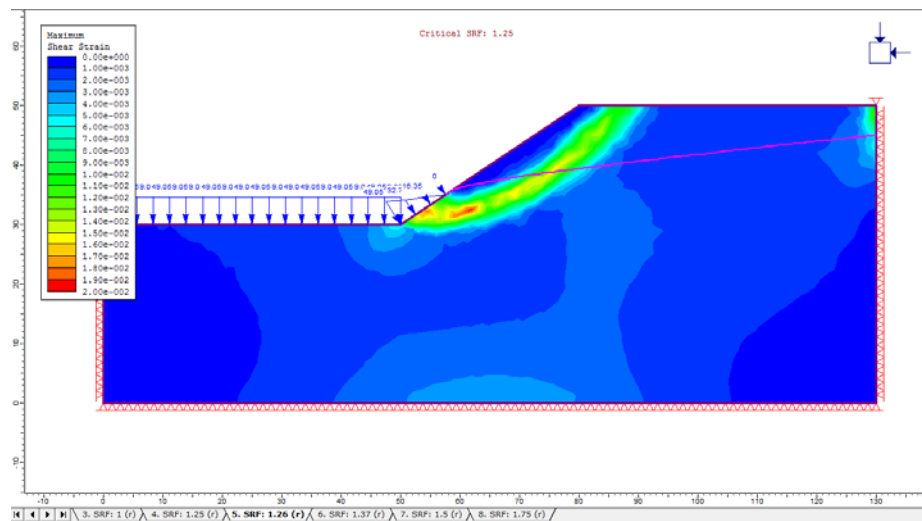
Interpret

The Interpret program starts and reads the results of the analysis. You are now looking at the maximum shear strain for the critical strength reduction factor (1.25) for the second stage.



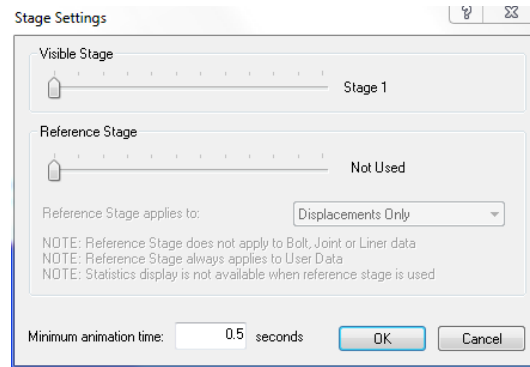
This is slightly lower than the SRF of 1.28 calculated for the model with Piezometric lines.

If you click the tab for SRF: 1.26 you get a better picture of the critical failure surface as shown.



This looks basically the same as the failure surface in the model with piezo lines.

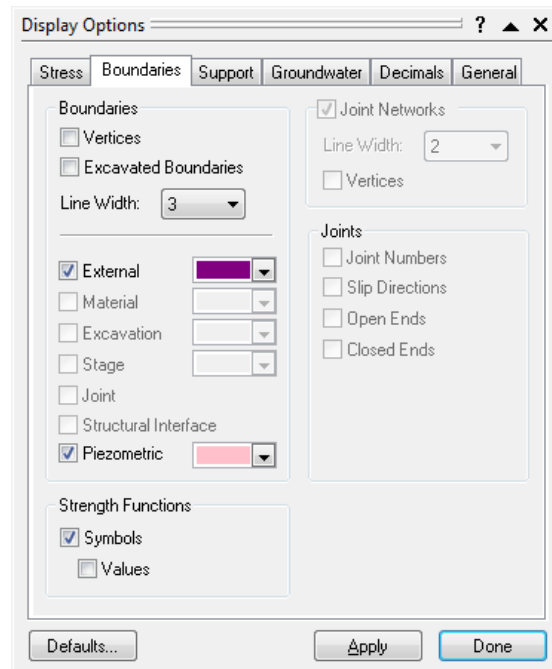
If you want to look at stage data prior to the SSR analysis, select **Stage Settings** from the **Data** menu. Set the Reference Stage to Not Used, and the Visible Stage to Stage 1.



Click OK. You will now see the maximum shear strain in Stage 1. Change the plot to Pore Pressure using the pull-down menu at the top. You can see the pore pressure in stage 1. Click the tab for Stage 2. It is clear how the pore pressure decreases as the boundary conditions change.

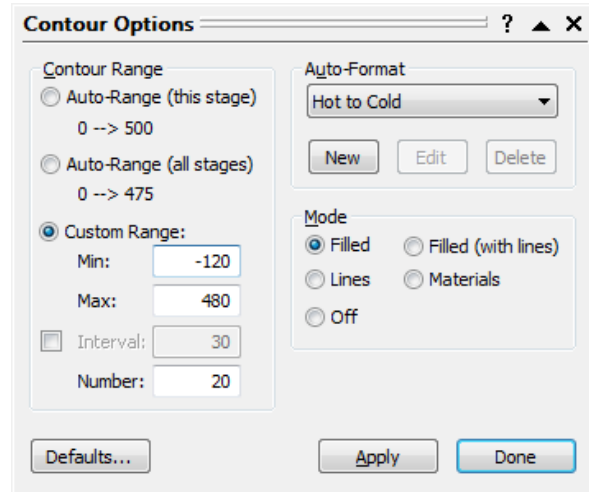
If you still have the previous model open in Interpret (with the piezo lines), you can view them both simultaneously by selecting Tile Vertically from the Window menu.

Click on the window showing the pore pressures in the piezo line model. Ensure you are looking at Stage 2. You can't see the piezo line because it is the same colour as the contours. To change the piezo line colour, select **Display Options** from the **View** menu. Under Boundaries, change the Piezometric line colour to pink as shown.

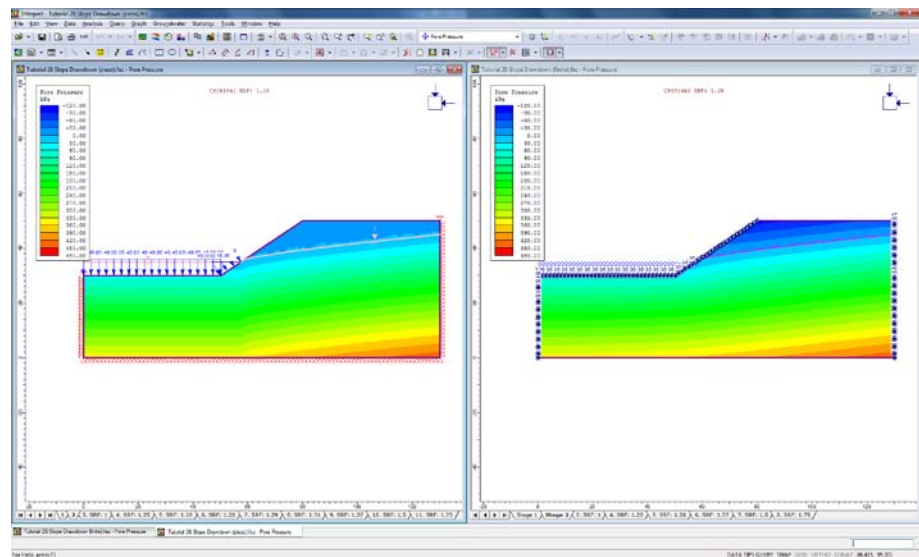


Click Done.

To facilitate comparison between the two models, we want the contour range to be the same. Select **Contour Options** from the **View** menu and select Custom Range. Set Min to -120 and Max to 480.



Click Done and the screen should look like this:



You will see the pore pressures are basically the same for the two models. The main difference is that the finite element groundwater model exhibits negative pore pressures (suction) above the water table.

Note that the negative pore pressures have no effect on the slope stability, unless you specify an unsaturated shear strength parameter ϕ_b for the material. See the *Phase2* help for more information.

The other difference between the models is that the contours are a bit smoother for the finite element groundwater analysis. The results from this model are likely more accurate than the results from the model with piezo lines. In the piezo line model, we had to guess at the water table profile and the water table was then used to compute pore pressures. In the finite element groundwater model, pore pressures are calculated based on the boundary conditions, and the water table shows where the pore pressure is 0.

This concludes the tutorial; you may now exit the *Phase2* Interpret and *Phase2* Model programs.